

Data Analytics Portfolio Global Warming Trends Analysis (1961–2022)

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Unit: MAN6777 Data Driven Managerial Decisions

Section 1: Managerial Problem Definition

Edith Cowan University (ECU) has an important strategic decision to make regarding long-term sustainability and the resilience of day-to-day operations. As the Sustainability Manager at ECU, the key decision focuses on the carbon reduction targets the university has, and on the investment the university is willing to put towards renewable energy infrastructure and climate resilience strategies by 2030, should the university fast-track such initiatives.

This decision carries a lot of strategic weight, more so now that areas around Perth have begun to experience the impact of climate change on a prolonged and heightened scale that makes day-to-day operations on the ECU campus even more challenging. That said, the cost to climate action compliance and adapting to the environment will get significantly greater the longer the ECU waits on these initiatives considering the changing regulations of the Australian government and the Tertiary Education Quality and Standards Agency (TEQSA).

Put aside the operational risks, climate change is the chief concern of students, international students especially, when it comes to choosing where to study. Climate leadership will enhance the university's brand and will help improve funding opportunities for research and partnerships with government and industry. Falling behind peer universities will have a negative impact on ECU in a competitive education environment. There's the potential of a long-term effect for risk and cost alignment with stakeholder expectations for a significant capital outlay. Budget challenges for public universities are ubiquitous; however, the cost/outcome 'gap' for not doing anything, in financial and reputational terms, is probably larger. Therefore, comprehensive and thorough analysis of the evidence of the historical and projected warming and its impacts will be invaluable to support the case for investment to the most senior levels and councils of the University.

Section 2: Data Analytics

The analysis is based on Global Warming Trends (1961–2022) dataset on Kaggle (Jawadawan, n.d.). This includes 11,222 records for 181 nations and includes the variables Country, ISO2, Year, and Temperature (with respect to the 1961–1990 baseline, temperature anomalies are in °C).

Data Preparation:

The dataset was downloaded via the kagglehub library and loaded into pandas. The year had an initial 'F' prefix which was removed, and the year was converted into an integer. Australia records were filtered into the dataset. Global temperature anomaly was calculated for each year, and was then smoothed with a 10 year rolling average, as was the Australian data, to show less volatility in the data and to highlight the longer term trends. After filtering, the dataset had no significant missing values.

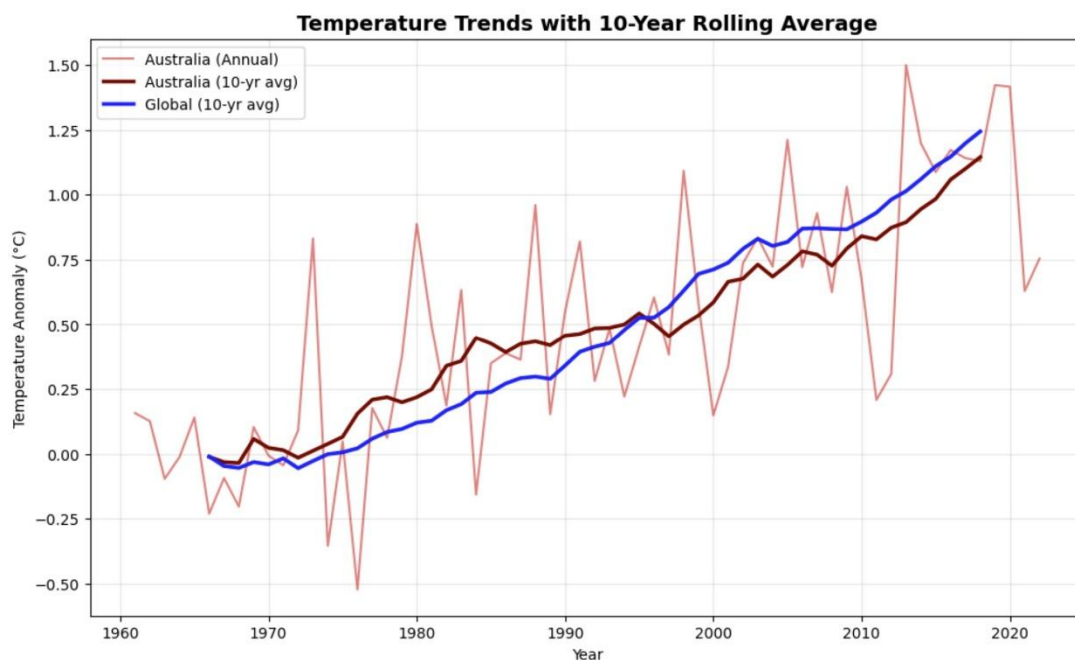
Analytical Techniques:

Descriptive statistics were used to understand central tendencies and variability. Time-series visualisation (line plots with rolling averages) was employed to observe trends visually. Linear Regression was applied to quantify the average warming rate ($^{\circ}\text{C}$ per decade) and assess the strength of the relationship using R^2 values. These techniques are appropriate because they effectively handle time-series climate data and provide clear, quantifiable insights to support long-term strategic decision-making.

Key Findings:

- Australia has experienced an average warming rate of **0.199°C per decade** ($R^2 = 0.568$).
- The global average warming rate is **0.250°C per decade** ($R^2 = 0.877$).
- Visual analysis reveals a clear upward trajectory in temperature anomalies, with noticeable acceleration in recent decades, particularly after 1990.

These results (see Figure 1: Temperature Trends with 10-Year Rolling Average) provide strong empirical evidence of sustained warming, which has direct implications for ECU's campus planning in Perth.



Trend plot with rolling averages

```
Australia data shape: (62, 4)
```

```
Global data shape: (62, 3)
```

```
Australia first 5 rows:
```

	Country	ISO2	Year	Temperature
9	Australia	AU	F1961	0.157
190	Australia	AU	F1962	0.126
371	Australia	AU	F1963	-0.096
552	Australia	AU	F1964	-0.012
733	Australia	AU	F1965	0.140

Data head / first 5 rows of Australia data

Section 3: Insights and Decision Implications

Recommendation:

By 2030, ECU should expedite its carbon reduction goals and allow for greater funding of renewable energy, particularly solar, and infrastructure that will support climate resilience.

Supporting Evidence:

Australia has an average increase of 0.199°C per decade as of 2022, compared to the global average of 0.250°C . Australia's trend line has a strong R^2 (0.877). Recent decades have shown that warming trends will lead to an increasing demand for cooling, therefore increasing operating costs and stressing the infrastructure of the already overburdened campus located in Perth. These results support the trends of global climate and the need for enhanced investment rather than periodic response.

Risks and Uncertainties:

Budgeting issues typical of the public sector may limit available capital for the build. The effects of climate change on ECU's campuses is not clearly defined because the dataset is limited to national and not localized to Perth. The future is uncertain with climate change and so are the rates of future warming.

Additional Information Needed:

To better inform decisions, ECU should obtain data on their campus energy use and temperature anomalies, and conduct detailed life-cycle cost-benefit analyses of solar and other energy infrastructure.

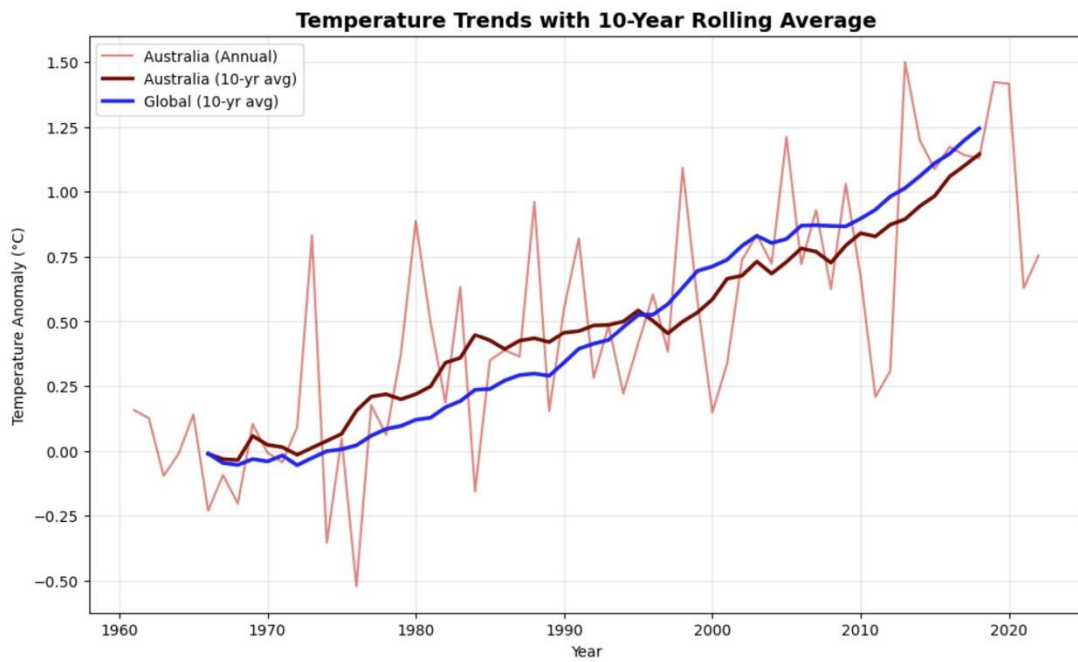
Final Action:

From 2026 to 2027, the first stage will include the implementation of solar and energy efficiency technology projects and phased over the next cycle will include the hard infrastructure to support technology. This balanced strategy allows ECU to manage financial risk while demonstrating environmental leadership.

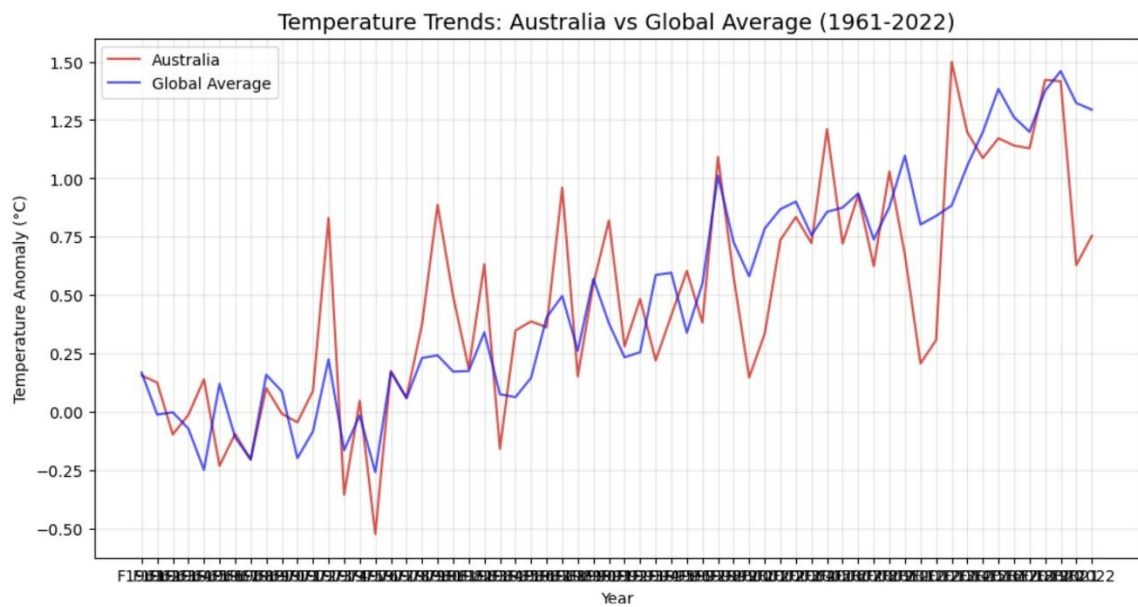
Data Analytics Portfolio – GitHub Workflow

Repository Link: <https://github.com/PaulinesoEcu/ecu-global-warming-analysis>

Key Visualizations:



1. Temperature Trends with 10-Year Rolling Average



2. Australia vs Global Trend

```
Year column cleaned successfully!
Sample years: [1961, 1962, 1963, 1964, 1965]

Australia Warming Rate: 0.199°C per decade
R² value: 0.568
Global Warming Rate: 0.250°C per decade
R² value: 0.877
```

3. Regression Output

```
Australia data shape: (62, 4)
Global data shape: (62, 3)
```

```
Australia first 5 rows:
```

	Country	ISO2	Year	Temperature
9	Australia	AU	F1961	0.157
190	Australia	AU	F1962	0.126
371	Australia	AU	F1963	-0.096
552	Australia	AU	F1964	-0.012
733	Australia	AU	F1965	0.140

4. Data Head (first 5 rows)

Main Python Code Summary (include 2–3 key code blocks you ran).

Reproducibility: All analysis done in Jupyter Notebook. Full code and data available in GitHub repository.